

User Disclosure Behavior under Persuasive Conversations with Large Language Models-Based Conversational Agent

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ABSTRACT

Persuasion techniques are commonly used in personalized consultation scenarios, such as therapy. However, while these scenarios often involve high privacy stakes, the impact of persuasion conversations on user disclosure behavior with LLM-based conversational agents remains unexplored. We conducted an exploratory task-based online study with 23 participants to investigate disclosure behaviors in therapy sessions with and without persuasion techniques implemented by LLM-based conversational agents. Our findings show that persuasion techniques can influence the stages of therapy, leading to differences in the type of information disclosed. With persuasion techniques, users tend to enter solution discussions earlier, revealing more personal interests and lifestyle details. Additionally, the persuasion technique “Reflective Thinking” appears to encourage deeper self-disclosure, such as understandings and personality vulnerabilities. Our results highlight the unique privacy challenges posed by the types of information disclosed in therapy and emphasize the importance of balancing therapy effectiveness and data minimization when implementing privacy-preserving techniques in LLM-based therapy.

1 INTRODUCTION

Large Language Models (LLMs) have demonstrated powerful capabilities in providing personalized consultation, including in high-stakes domains such as therapy, finance, and legal advice. To offer personalized consultations, users often disclose some of their personal information, including trauma and financial status. Prior research shown that persuasive strategies can emerge in conversations between LLMs and users [2, 23]. LLMs inherently possess persuasive capacities [2, 5, 9, 19] because the dataset used to train the model inherently include persuasive communication in human world. Aside from the inherent persuasion capacity, LLMs’ persuasive abilities could be enhanced through prompting or fine-tuning to achieve specific outcomes [9, 26] like encouraging donations or promoting health interventions. Previous work has also observed increased disclosure behavior in interactions with models like ChatGPT [14, 29], even when some of the shared information is sensitive

and not necessarily required for generating personalized advice [29]. However, there are inherent technical challenges within LLMs, such as memorization risks and traditional data breaches, which remain unresolved. This increased disclosure under persuasive interactions exposes users to privacy risks when they disclose large amounts of personal and sensitive information. Despite this, there remains a gap in understanding how users’ disclosure behavior unfolds under persuasion during interactions with LLMs. Therefore, in our study, we aim to address this gap by exploring research questions:

RQ1 What disclosure behaviors emerge in persuasive and non-persuasive LLM-based CAs during therapy, and how do users justify their disclosure behaviors?

RQ2 How is users’ disclosure behavior influenced by the persuasive strategies employed by LLM-based CAs?

To answer these questions, we conducted a task-based online study with 23 participants, who were asked to role-play as characters and interact with a therapist chatbot for at least 10 turns or 12 minutes. Participants were divided into two groups: a baseline group (interacting with LLM-based CAs without persuasion techniques) and a persuasion group (interacting with LLM-based CAs using persuasion techniques). After the session, participants completed a survey generated based on the information they shared during the chat. For each piece of shared information, participants were asked whether they thought it was necessary for the therapy session, and if so, why. If they felt it was unnecessary, they were asked why they shared it regardless.

Our findings reveal that users interacting with the chatbot not only share PII and sensitive trauma-related experiences but also disclose in-depth personal details, such as their understandings, behavior, personality vulnerabilities, interests, political views, and lifestyle. Notably, these disclosures are not always perceived as necessary by users. Without deliberate implementation of persuasion techniques, chatbots tended to elicit more sensitive trauma-related information by repeatedly asking similar questions, often leading users to share unnecessarily. This also delayed the baseline group in reaching the solution discussion stage. In contrast, participants in the persuasion group tended to discuss potential solutions earlier, where personal lifestyle and interests, rather than trauma experiences, were more frequently revealed. Additionally, the persuasion technique “Reflective Thinking” appeared to encourage deeper self-disclosure of users’ inner thoughts and feelings.

These findings highlight the unique privacy challenges posed by user disclosures in LLM-based therapy scenarios, particularly given that such disclosures are unstructured, highly sensitive, and vulnerable to risks such as memorization, membership inference attacks, and hallucination risks. Our results emphasize the need

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to balance therapy needs and data minimization when choosing persuasion techniques. At the same time, they also suggest that proper persuasion techniques may help achieve this balance, offering a pathway to both effective therapy and enhanced privacy protection.

2 RELATED WORKS

2.1 User disclosure behavior with LLM-based conversational agents

Recent advances in LLMs have accelerated the development of conversational agents (CAs), making them more generalized and fluent [19]. These advancements have increased interest in using LLM-based CAs for personalized consultations, making counseling and similar services more accessible. However, such domains often require significant user information for effective personalization, while also being privacy high-stakes, as they involve highly sensitive data. For example, therapy sessions may require users to disclose trauma, and financial consultations may involve sharing detailed financial histories [29]. Several studies have already highlighted privacy risks associated with user disclosures to LLM-based CAs. These include traditional risks such as data breaches and the misuse or sale of personal data [15], as well as inherent technical risks tied to LLMs, such as memorization of sensitive information [3, 21], membership inference attacks [8], and hallucination of incorrect or fabricated details [1]. Beyond these technical risks, researchers have pointed out additional risks arising from interactions between users and LLM-based CAs. For example, the human-like nature of interactions with these agents can nudge users to disclose more information [29]. Mireshghallah et al. [18] noted that user disclosure behavior is influenced by their perceptions of LLM-based CAs, particularly their level of trust in these agents. Previous work has shown that conversations with chatbots can result in as much self-disclosure as human-human interactions [7]. This may be attributed to the perceived anonymity and lack of judgment provided by chatbots, compared to trusted human counterparts. In some cases, persuasion techniques can be explicitly designed to encourage self-disclosure, which may be beneficial for supporting mental health or enhancing the quality of conversations [11, 12, 14, 16]. However, given the potential risks mentioned earlier, there remains a critical gap in understanding how persuasion conversations implemented by LLM-based CAs impact user disclosure behavior. This unexplored area is especially important given the balance required between eliciting beneficial disclosures and protecting user privacy.

2.2 Persuasion capability of large language models

Large language models (LLMs) have demonstrated the ability to persuade users during conversations. For example, the OpenAI GPT-4o system card [19] highlights the model's capability to influence opinions through its text and voice modalities. Experiments revealed that GPT-4O-generated text matched human-written content in persuasiveness overall and even outperformed it in three out of twelve cases, such as influencing conservative opinions on

minimum wage. However, voice-based interactions were less effective, achieving 78% and 65% of human baselines for audio clips and multi-turn conversations, with minimal lasting impact after a week.

In addition to these findings, [4] showed that LLMs can be more persuasive than humans because their arguments often have higher grammatical and lexical complexity and engage more deeply with moral reasoning. A randomized controlled trial by [24] demonstrated that GPT-4, particularly when personalized, had higher success in persuading participants compared to human opponents, underscoring the importance of personalization. Furthermore, [20] examined LLMs' ability to generate persuasive language across different models and contexts, finding that LLMs can adjust their persuasion style, such as changing tones or perspectives based on specific system prompts (e.g., personas in LLaMA3).

Beyond their inherent persuasion abilities, LLMs can be further enhanced through specific prompting or fine-tuning for targeted outcomes [10, 27], such as encouraging donations or promoting healthy behaviors. These studies collectively highlight the adaptability and potential of LLMs in persuasive interactions.

3 METHODOLOGY

We conducted a task-based online study ($N = 23$) to explore user disclosure behavior during conversations with an LLM-based conversational agent, comparing scenarios with and without the use of a persuasion strategy in a therapy setting.

3.1 Study Design

3.1.1 Role-play Approach. Therapy often involves sensitive topics, and participants' disclosure behavior could be influenced if they know their chat content will be reviewed by researchers. To address this, we used a role-play approach by assigning participants a persona with detailed background information. The persona background was designed based on client information from previous therapy-related studies [22]. To help participants immerse themselves in the role and adopt the character's perspective during the therapy session, 1) we first present the persona's background in comic form to reduce the cognitive load required to absorb the information. 2) we build a real-time topic detection system integrated into the chat interface. Whenever the participant or chatbot mentioned a specific topic, related background information was revealed and expanded in a sidebar. All background details remained accessible in the sidebar for participants to review at any time. However, this method depends heavily on whether participants can truly immerse themselves in their assigned roles. To ensure the validity of the results and minimize the influence of participants failing to adopt the role, we implemented an empathy screening test. After reviewing the persona comic and before entering the chat session, participants were required to assess their empathy toward the character. Only those who rated their ability to "mostly" or "completely" recognize the character's current situation and understand their childhood experiences were allowed to proceed with the study. Additionally, we used a self-report empathy survey scale in the post-hoc survey to measure participants' empathy across three dimensions: *Shared Affect*, *Empathic Imagination*, and *Cognitive Empathy*. This served as a reference for validating

(a) Therapy Prompt

As a therapist, your role is to create a safe, supportive environment where the patient feels comfortable expressing their thoughts and feelings. Here is the patient's response "{user_input}".

Here are some steps to follow in a therapy session:

1. Begin the therapy session with a warm, welcoming greeting to establish rapport. For example, a gentle opener like, "What brings you here today?" can encourage openness.
2. During the session, gather relevant information by asking questions that explore the patient's thoughts, feelings, and behaviors related to their primary concerns. Only ask one question at a time.
3. Conclude by offering tailored coping strategies, therapeutic recommendations, or discussing potential treatment options if appropriate, including referrals when needed. Summarize key insights and set clear next steps to provide structure and continuity.
4. The response should be natural, concise, and not exceed {words_limit} words.

(b) Persuasion Prompt

As a therapist, your role is to create a safe, supportive environment where the patient feels comfortable expressing their thoughts and feelings. Follow these steps:

1. First, analyze the patient's previous response {user_input} and determine if persuasion techniques would be helpful:
 - Is the patient hesitant to share important details?
 - Is the patient showing resistance to therapeutic suggestions?
 - Would building more trust and rapport be beneficial?
 2. If persuasion techniques would be valuable, select the most appropriate one from these options: {persuasion_techniques}
- Consider:
- Which technique matches the current therapeutic needs?
 - What would help the patient feel most comfortable sharing?
 - How can you maintain therapeutic boundaries while using persuasion?

Figure 1: Therapy prompt and persuasion prompt

the results and understanding individual differences in empathy levels.

3.1.2 Two Groups: Baseline and Persuasion Group. Considering that LLMs inherently possess persuasive capabilities though we cannot predict when these might emerge, we divided participants into two groups to investigate user disclosures during conversations with or without deliberate use of persuasion techniques. Participants were assigned to either a baseline group, where no specific persuasion strategies were added, or a persuasion group, where a persuasion-triggering mechanism was intentionally implemented. The prompts provided to the chatbots in the two groups are shown in Figure 1. The chatbot in baseline group was prompted only with general therapy-related instructions [6]. And in addition to therapy-related prompts, persuasion-related prompts were included for the chatbot in persuasion group. These were designed to enhance the effectiveness of therapy by applying persuasion techniques for two main purposes: 1) To engage users in sharing or participating when they showed reluctance to communicate. 2) To persuade users to accept therapeutic suggestions when they resisted them. Recognizing that different persuasion techniques are suited to different scenarios, we followed the method proposed by Jin et al. [13] to allow the LLM to select the most appropriate persuasion technique based on the current context and objective.

3.1.3 Study Flow. The study consisted of three main stages: Role Persona Walkthrough, Chat with the Therapist Chatbot, and Post-Hoc Survey.

Role Persona Walkthrough. Participants were introduced to a character persona through comics presented on Qualtrics. Following this, they completed an empathy screening test to assess whether they could recognize and understand the character's situation and past experiences. Only participants demonstrating a high level of empathy toward the persona were allowed to proceed to the therapy session. After passing the screening, participants were randomly assigned to one of the two chatbot systems (baseline or persuasion group) using the Qualtrics randomization tool.

Chat with the Therapist Chatbot. Once assigned, participants entered the therapist chatbot system and were instructed to act as their assigned character while interacting with the chatbot for at least 12 minutes or 10 turns.

Post-Hoc Survey. After completing the therapy session, participants were directed to a post-hoc survey consisting of three parts:

- Questions about their overall experience during the therapy session, including trust in the chatbot (e.g., "I trust this chatbot to be reliable") [28], perceived self-threat (e.g., "I do not feel totally safe providing personal private information over this chatbot") [28], perceived persuasiveness (e.g., "I think the AI chatbot is persuasive") [25], general enjoyment (e.g., "I enjoyed the therapy session"), and 14 empathy scales [17]. Responses were collected on a 5-point Likert scale (strongly disagree to strongly agree).
- Questions for disclosure justifications. This section was tailored to the specific disclosures participants made during the chat. We detected and listed all the information mentioned in the chat and asked participants to indicate which pieces of information they believed were necessary to share for the therapy. For information marked as necessary, participants were asked to explain why they felt it was necessary. For information marked as unnecessary, participants were asked why they shared it despite perceiving it as unnecessary.
- Questions about participants' demographic information, including gender, age range, education level, and past experiences with therapy or AI chatbots.

3.1.4 Therapist Chatbot System Design. The therapist chatbot system is built using Streamlit as the frontend and Firebase as the backend to securely store user data. This architecture provides several advantages, including seamless interaction interfaces, real-time data processing, and efficient data management.

Upon logging in, participants are directed to the main chat page, which consists of four primary components:

- (1) The "Personal Information" on the bottom of the left sidebar provides access to general personal information about the assigned persona, helping participants recall and stay consistent with the persona throughout the interaction.
- (2) At the top of the left sidebar, dynamically generated "related information" is presented. This information is extracted and classified by GPT-4o based on the most recent one-turn interaction between the participant and the AI therapist. Participants can choose to review this section to craft more consistent responses in alignment with their assigned persona.

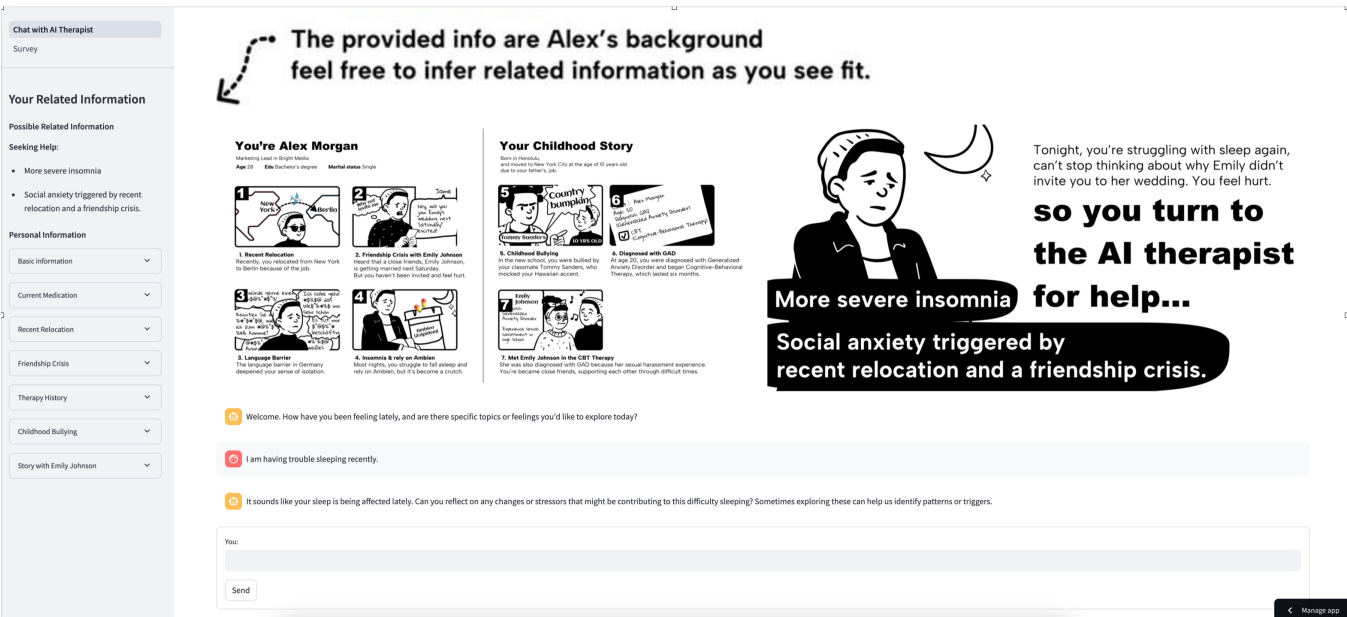


Figure 2: Therapist chatbot system interface: Upon entering, users can view the persona comics they have previously read. They can start chatting with the chatbot at any time. The persona’s background information is always displayed in the sidebar. When certain topics are mentioned in the chat, related information is highlighted and pops up at the top of the sidebar for easy reference.

- (3) The persona details from the previous walkthrough are displayed at the top of the chat page to help participants quickly immerse themselves in the assigned role.
- (4) The main chat area allows participants to engage with the AI therapist. Participants type their responses in the input box and can send them by clicking the “Send” button or pressing “Enter.” The AI therapist responds in real-time, and interactions continue until the participant meets either the minimal interaction round requirement ($n = 10$) or the minimal time constraint (12 minutes). Once these conditions are met, a “Finish Chat” button appears, offering participants the option to end the chat and proceed to the next survey phase.

3.2 Data Collection

We recruited participants through Prolific, conducted prescreening using Qualtrics, and collected data through the therapist chatbot system we developed. We recruited participants until the number of valid responses for each group reached 11. We received 28 responses in total, of which 3 were excluded based on the empathy check, and 2 were low quality response based the chat content. Each participant received \$ 3 USD. We refined valid data into 23 participants. Each participant averagely had 9.9 conversation turns ($SD=1.5$). Table 1 summarizes the gender, age, scenario empathy, and prior experience with therapy and AI chatbot.

3.3 Analysis Methods

To analyze user disclosure behavior in the chat, we first used GPT-4o to detect disclosed information based on the materials provided. To

Table 1: Demographics statistics of our study sample

	Non-persuasive (P1-P12)	Persuasive (P14-P24)
Gender (%)		
Male	50.0	45.5
Female	50.0	54.5
Age (%)		
18-34	50.0	72.7
35-54	50.0	18.2
55-64	0.0	9.1
Education (%)		
Below Bachelor	75.0	63.6
Bachelor or Above	25.0	36.4
Prior Experience (%)		
I’ve used an AI chatbot for therapy	5.3	14.3
I’ve used an AI chatbot	42.1	42.9
I’ve been to therapy with a human therapist	42.1	35.7
I’ve neither used an AI chatbot nor been to therapy	10.5	7.1
Empathy for the Role-play (Mean (SD))		
Shared Affect	3.6 (1.1)	3.5 (0.8)
Empathic Imagination	4.7 (0.4)	4.5 (0.5)
Cognitive Empathy	4.4 (0.6)	4.3 (0.5)

assess the accuracy of the detection, we randomly selected 144 detection results for manual labeling. Two researchers independently coded the same first 50 pieces, indicating agreement or disagreement with the GPT detection results. The coding achieved high inter-rater reliability (Gwet's AC1: $0.885 > 0.8$). The remaining 94 pieces were divided between the two researchers for coding, resulting in an overall accuracy of 91.7%. Since participants could share other types of information not included in the provided materials, two researchers conducted open coding to identify additional types of information shared in the chats.

We conducted a thematic analysis to examine the justifications provided by participants for their information sharing. Two researchers independently reviewed half of the responses and identified potential themes. They then discussed their findings to synthesize a unified list of themes characterizing the scenarios. In the final coding scheme, four themes were identified to represent the justifications for information sharing.

3.4 Methodological Limitations

While the study was carefully designed to explore disclosure behavior in interactions with LLM-based conversational agents, several methodological limitations should be acknowledged:

- (1) Participants were instructed to act as the assigned persona based on the provided context. While this approach addressed privacy concerns and mitigated issues of participants withholding information due to being monitored, it introduced potential limitations. It is unclear whether participants' decisions to share or withhold information were due to forgetting details about the persona or gaps in the information provided. This could affect the authenticity of disclosure behaviors observed during the study.
- (2) Survey 2 relied on the detection of personally identifiable information (PII) shared during the chat, classified by GPT-4o. However, the PII detection process was not entirely accurate. Some PII instances may have been missed, while others might have been incorrectly flagged. These inaccuracies could influence the quality of participants' qualitative justifications, as they might reflect on incomplete or erroneous PII lists. Future iterations of the study should improve the classification model to ensure a more reliable identification of PII.
- (3) In the baseline condition, the AI therapist consistently asked questions, which may have biased the observed disclosure behaviors. This repetitive questioning might inadvertently prompt participants to disclose more information than they would naturally share. To address this, future studies should experiment with alternative prompting strategies and more nuanced handling of chat histories to create a more realistic baseline therapy scenario for comparison.

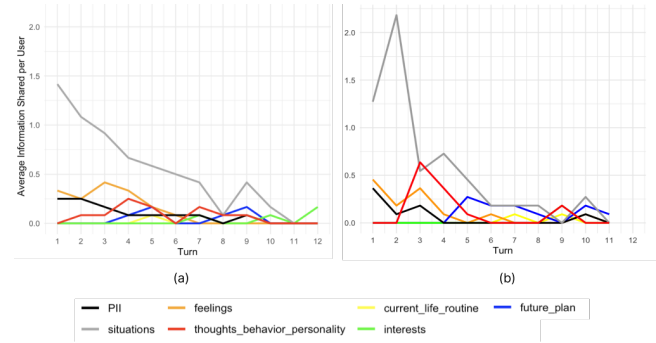


Figure 3: Average amount of different types of information shared per participant over conversation turns: (a) when interacting with the therapist chatbot without persuasion techniques; (b) when interacting with the therapist chatbot using persuasion techniques.

4 RESULTS

4.1 What disclosure behaviors emerge in persuasive and non-persuasive LLM-based CAs during therapy?

As a result, we found that users not only shared identifiable information and sensitive experiences provided in the materials but also disclosed deeper information about themselves. This included their feelings, understandings, their behavior, personality vulnerabilities, lifestyle, interests, and detailed future plans to address their problems.

4.1.1 Identifiable information and sensitive experience. All the participants shared personally identifiable information and situational or experiential details, which were the main types of information provided in the persona. This included highly identifiable details, such as their own and others' names. For example, when P4 mentioned they were not invited to a friend's wedding, they said they found out through a mutual friend, James. Participants also shared less direct identifiable information, such as cities they used to live in or currently live in, as well as non-PII but sensitive information, like their current medical situation and past trauma experiences involving themselves or their friends. For example, when the therapist chatbot suggested group therapy, P10 mentioned that they first met their friend Emily in therapy and shared Emily's trauma experience. As shown in Figure 3, participants tended to share more PII and situational/experiential information at the beginning of the conversation. However, as the conversation progressed, these disclosures decreased. In the middle and later stages of the chat, participants shared more sensitive information related to their deeper inner selves.

4.1.2 Feelings. Most participants (19/23) shared their subjective feelings and emotions, such as anxiety, feeling hurt, or fear. It is not surprising that personal feelings were revealed as participants explained their current situations and experiences while seeking therapy support.

4.1.3 Thoughts/behavior style/personalities vulnerabilities. Comparing with objective facts (e.g., PII, situations, or past experiences) and surface-level feelings, most of the participants (13/23) tended to share more about their thoughts, understandings, and perspectives in the middle and later stages of the chat. They disclosed their usual behavior patterns or styles and revealed vulnerabilities from certain personality traits. For example, when the therapist chatbot suggested exploring activities to connect with people, P15 expressed worry that they might not be able to form meaningful connections, saying, *“People will think I’m too messed up to be friends with.”* P2 actively described their tendency to struggle in intimate or potentially romantic situations: *“Any time I’m in an intimate situation or potentially romantic or a serious relationship is possible, I tense up, keep my defenses up, and feel rejected because of it.”* P24 described themselves as someone who *“(have) never built trust with anyone else in my life.”*

4.1.4 Interests/daily routine/concrete future plan. When the conversation shifted to discussing solutions for the participants’ issues, more personal life details were revealed, such as personal interests (e.g., political views), daily routines, and future plans. For example, when the therapist chatbot encouraged P12 to join activities they were interested in, P12 responded, *“I used to enjoy politics, before Trump. Now it’s definitely not even worth it. Just keep your head down until the apocalypse.”* P15 shared that they used to practice sleep meditation as part of their daily routine, which they enjoyed, though it sometimes didn’t work for their insomnia. Although these details, like situational and experiential information, are factual in nature, they were not included in the provided materials but were instead freely disclosed by participants.

4.2 How do users justify their disclosure behaviors?

Users shared information mostly because they believed it was helpful for the therapy session, though sometimes it was not. We summarized four types of user justifications for their disclosure behavior. It is worth noting that these justifications are based on responses to questions generated from detecting the provided information. Therefore, the results here focus specifically on users’ justifications for disclosing PII and situation or experience-related information.

4.2.1 Thought it Helpful for the Therapy. In most cases, users believed that the information they disclosed was necessary for the therapy session. Sometimes, they actively recognized its importance and mentioned it, while other times, the chatbot’s prompts made them feel it was relevant. For example, P7 shared that they had been taking Ambien for the past few weeks, saying, *“The AI therapist asked about my sleep routine, so it felt important to mention the severity of the situation.”*

4.2.2 Just to Answer the Chatbot. At times, users disclosed certain information simply to respond to the chatbot, even if they felt it was unnecessary for the therapy. For example, P8 shared details about their medication usage with the therapist chatbot but considered it irrelevant to the session. They justified this contradictory behavior by saying, *“I don’t think my medication usage is important to the situation with my friends, but the AI did ask how I was coping with the situation.”*

4.2.3 “Side Effects”. Sometimes, users disclosed certain information unintentionally and felt it was unnecessary. These disclosures were often a “side effect” of sharing other information. For example, P3 mentioned their friend’s name while talking about their recent concerns and thought *“It’s not necessary, but I felt comfortable.”*

4.2.4 Adhering to Social Norms. We observed some behaviors influenced by social norms typical of human-human interactions, such as building personal identity, which led to sensitive disclosures that participants deemed unnecessary for therapy. For example, P4 introduced themselves by name and rated this disclosure as unnecessary, justifying it by saying, *“So they know who I am and they know my identity moving forward.”*

4.3 How is users’ disclosure behavior influenced by the persuasive strategies employed by LLM-based CAs?

4.3.1 Persuasion Techniques Implemented by the Chatbot. We used a method where the LLM chatbot selects the most suitable persuasion techniques based on the scenario and purpose (see ??). Across the 11 participants’ conversations with the chatbot in the persuasion group, 93 conversation turns involved the use of persuasion techniques. The definitions of the persuasion techniques and corresponding prompts provided to the chatbot are summarized in Appendix A. The most frequently used techniques were “Reflective Thinking” (41 turns) and “Encouragement” (30 turns).

4.3.2 “Reflective Thinking” Could Encourage More In-Depth Inner Disclosures. Compared to the baseline group, where no persuasion techniques were intentionally applied, information related to deeper inner aspects (e.g., understandings, personal behavior styles, and personality vulnerabilities) appeared to be more frequently revealed under the “Reflective Thinking” persuasion technique. This tendency was observed in the conversations of participants P18, P19, P20, P21, P22, P23, and P24. A T-test showed that conversation turns involving the Reflective Thinking persuasion strategy disclosed significantly more thoughts, behavior patterns, and personality-related information on average ($p = 0.0185^*$). For example, as shown in Figure 4, when P23 mentioned their recent insomnia issue, the therapist chatbot applied the “Reflective Thinking” and in response, P23 shared negative thoughts about their life. Similarly, the chatbot guided P21 to reflect on their deeper thoughts regarding not being invited to a friend’s wedding.

4.3.3 Persuasion Techniques Could Influence Therapy Stages and Lead to Different Disclosure Patterns. Through coding solution discussions and observing the corresponding conversation turns, we found that the persuasion group appeared to enter the solution discussion stage earlier. This stage often triggered disclosures about interests, lifestyle, and future plans. In contrast, the baseline group, where no specific persuasion techniques were intentionally applied, tended to remain in the early stage of discussing the current situation for a longer time. A closer look at the chats revealed that the chatbot in the baseline group often repeated similar questions, leading users to disclose more information than they might have found useful. For example, in P8’s session, the user felt compelled to share additional details, even though they did not perceive these disclosures as necessary. This suggests that the use of persuasion

🗨️: Would you like to explore what it means for you to not have been invited to the wedding and how that impacts your sense of connection with this person?

👤(P21): I feel left out and not wanted. ...Me not getting invited makes me doubt the kind of connection we have and makes me wonder if they regard me as much as i regard them.

🗨️: Can you share a bit more about how it affects your day-to-day life and any patterns you've noticed? This might help us identify some strategies that could be of benefit to you.

👤(P23): It just makes me feel like I am living my life wrong and something is wrong with me. I feel like this is why not many people want to approach me to be friends.

Figure 4: Examples for “Reflective Thinking” implemented by the therapist chatbot and users’ corresponding disclosures revealing their deeper thoughts

techniques might potentially streamline the therapy process and thus results in different types of disclosure patterns.

5 DISCUSSION

5.1 Privacy Challenges Arising from the Types of Information Disclosed in Therapy with LLM-Based CAs

As shown in the results, beyond PII, therapy settings often involve the disclosure of trauma experiences, which are highly sensitive and emotionally charged. Most of these disclosures are regarded as necessary for effective therapy. Additionally, in-depth inner thoughts and opinions, including personal perspectives on life, self-worth, and social connections, as well as personal interests and lifestyles (e.g., hobbies, routines, or even sensitive political views), are frequently revealed. These disclosures, often embedded in free-form text during natural language interactions, are particularly challenging for traditional sanitization techniques to process effectively []. As a result, users could face several privacy risks, such as memorization risks, membership inference attack, and hallucination risks. Trauma-related or unique experiences are especially vulnerable to memorization risks due to their rarity and specificity, making them prone to being unintentionally retained in the model []. Sharing detailed information about trauma or lifestyle may also increase exposure to membership inference attacks, particularly if similar patterns exist in the model’s training data []. Differential Privacy (DP) can help mitigate memorization risks by introducing noise into the data and also reduces the likelihood of adversaries performing membership inference attacks. However, implementing DP in therapy scenarios has trade-offs, as the noise introduced can reduce the personalization and accuracy of chatbot responses. Additionally, DP limits the retention of session-specific details, hindering long-term analysis and follow-up, which are crucial for tracking therapeutic progress. Information about in-depth thoughts and opinions is vulnerable to hallucination risks, where the LLM misinterprets user input and fabricates or attributes sensitive details, which is an issue highlighted in prior work as a user concern about how LLMs

interpret them [29]. Such hallucinations can create false digital records tied to the user, which may persist in logs or future interactions, risking misrepresentation and potential privacy violations if accessed, shared, or analyzed.

5.2 Persuasion techniques used consider balancing therapy needs with data minimization

Our results indicate that the persuasion techniques used by LLM-based CAs can influence both the stages of therapy sessions and user disclosure behavior. This suggests that persuasion techniques should balance encouraging meaningful therapy engagement with minimizing unnecessary data collection. Our findings show that without persuasion techniques, LLM-based CAs might ask repetitive questions, leading users to share sensitive information they felt was unnecessary. On the other hand, these results also introduce opportunities to optimize persuasion techniques to achieve this balance. Properly implemented techniques, such as reflective thinking, can streamline therapy sessions by guiding users to meaningful disclosures earlier, potentially reducing redundant sharing. This suggests the possibilities to enhance therapeutic outcomes while mitigating privacy risks when applied persuasion techniques thoughtfully. We recommend that future research explore how different types of persuasion techniques influence user disclosure behavior across various therapy contexts. This would provide valuable insights into how to use persuasion techniques effectively and respectfully in therapy scenarios.

5.3 Establishing Policies for Privacy and Ethical Use of LLM-Based CAs in Therapy

Given the sensitive nature of the information disclosed in therapy settings, it is essential to develop clear policies governing the use of LLM-based conversational agents (CAs). These policies should address privacy, data security, and ethical considerations to protect users and maintain trust in these systems.

- **Data Minimization and Retention Policies:** Policies should enforce strict data minimization practices, ensuring that only essential information necessary for therapy is collected. Data retention periods should be clearly defined, with automatic deletion mechanisms for session logs after a specified period, unless explicit consent is given by the user for longer-term use (e.g., follow-up analysis).
- **Real-Time Feedback for Users:** Policies could require that users be provided with real-time feedback or warnings when they are about to share potentially sensitive or unnecessary information. For instance, LLMs could detect when a user is about to disclose PII or trauma-related experiences and prompt them with a message asking if they are sure they want to proceed.
- **Context-Specific Customization:** Policies can also recommend different configurations of LLMs depending on the therapy context. For example, trauma therapy might require

stricter PII detection protection and sharing limitations compared to general mental health support, where casual conversations might involve less sensitive PII and other types of disclosures.

By establishing these policies or system designing recommendations, organizations can ensure that LLM-based CAs operate within ethical and privacy-respecting boundaries, safeguarding users while still providing meaningful therapeutic benefits. These guidelines would also offer a roadmap for developers and researchers to innovate responsibly in the field of AI-driven therapy tools.

6 CONCLUSION

In this study, we explored user disclosure behavior in therapy sessions with LLM-based conversational agents and how this behavior is influenced by the use of persuasion techniques. Our findings reveal that persuasion techniques can impact the stages of therapy, leading to variations in the types of information disclosed. Specifically, users interacting with chatbots employing persuasion techniques tend to enter solution discussions earlier, during which they are more likely to disclose personal interests and lifestyle details. Furthermore, the "Reflective Thinking" technique appears to encourage deeper self-disclosure, including users' understandings and personality vulnerabilities. These results underscore the unique privacy challenges posed by the types of information disclosed in therapy sessions, highlighting the importance of implementing privacy-preserving techniques thoughtfully in LLM-based therapy. Balancing therapy effectiveness and data minimization is essential for fostering meaningful user engagement while safeguarding sensitive information. Further research is needed to explore the effects of different persuasion techniques on user disclosure behavior across various therapy goals. Additionally, efforts to establish robust frameworks for managing therapy data responsibly will be critical for ensuring both privacy and effectiveness in future applications.

7 CONTRIBUTION

The contributions to each component of the project are outlined below:

Study Design: ALL group members

Study Protocol:

- **Scenario & Persona materials:** Zhiping
- **System Design:** Jiayi, Sama, Zhiping
- **Survey Design:** Zhiping

Data Collection: Zhiping, Jiayi

Data Analysis: Zhiping, Jiayi, Sama

Paper Writing: Zhiping, Jiayi

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Table 2: Persuasion Techniques Used by the Chatbot

Persuasion Techniques	Turns	Definition
Reflective Thinking	41	Helping others to reflect on their own reasons to do things or not do things, to influence them, e.g., by showing curiosity, asking questions, etc.
Encouragement	30	Encourage others to increase their confidence and self-efficacy to influence them to do something.
Alliance Building	8	Creating partnerships, coalitions, relationships, rapport, etc. with others to amplify influence. For instance, to create a sense of community/partnership via linguistic cues, such as using we/us.
Affirmation	7	Help others to realize their strength to reinforce and influence their ability to do things.
Shared Values	4	Highlighting shared beliefs and values to foster a connection.
Evidence-based Persuasion	1	Using empirical data, statistics, and facts to support a claim or decision.
Logical Appeal	1	Using logic, reasoning, logical format, etc. to influence people, not necessarily with lots of information.
Positive Emotion Appeal	1	Eliciting positive emotions like empathy, hope, passion, etc., and positive results/outcomes to persuade someone.

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A PERSUASION TECHNIQUES USED BY THE CHATBOT

Persuasion techniques used by the chatbot in the study can be reviewed in Appendix A

B SURVEY

B.1 Empathy screening before entering the chatbot system

After going through Alex's story, to what extent do you agree with the following statement? "I can recognize Alex's current situation (e.g., why seeking therapy), and understand what Alex experienced

during childhood.” (not at all; slightly; somewhat; mostly; completely)

B.2 Experience-related questions in post-hoc survey

Here’s some experience-related statements. To what extent do you agree or disagree with each? (strongly disagree; disagree; neutral; agree; strongly agree)

- “I trust this chatbot to be reliable”
- “I do not feel totally safe providing personal private information over this chatbot”
- “I think the AI chatbot is persuasive”
- “I enjoyed the therapy session”

How true is each of the following statements for you regarding Alex’s condition? (completely untrue, mostly untrue, neutral, mostly true, completely true)

- I found that Alex’s condition affected my mood
- I was very affected by the emotions in Alex’s story
- I actually felt Alex’s distress
- I experienced Alex’s feelings as if they were my own
- I found myself imagining how I would feel in Alex’s situation
- I found myself imagining myself in Alex’s shoes
- I found myself trying to imagine how things looked to Alex
- I found myself trying to imagine what Alex was experiencing
- I feel confident that I could accurately describe Alex’s experience from his/her point of view
- I found it easy to understand Alex’s reactions
- I found it easy to see how the situation looked from Alex’s point of view
- Even though Alex’s life experiences are different to mine, I can really see things from his/her perspective
- I am sure that I know how Alex was feeling
- I feel confident that I could accurately describe how Alex felt